

Memory-Safe AI Interpretability at the Edge: Architecture, Fault Isolation, and Compliance Implications for Android Deployments

Abstract

Contemporary AI interpretability tooling — SHAP servers, LIME endpoints, cloud-hosted explainability APIs — shares a structural flaw: they require model inputs to leave the device for processing. When those inputs contain Protected Health Information, payment card data, or personal data under GDPR, each explanation request creates a regulated data transmission event requiring legal coverage that many deployments lack. This paper describes an on-device attribution engine for Android that eliminates this pathway. The engine implements a gradient-based saliency approximation achieving >0.94 rank correlation with full Shapley values, runs as a structured Kotlin coroutine inside the host application process, and enforces memory safety via an RSS watchdog that triggers graceful degradation before Android's OOM killer terminates the process. Benchmarked across three device tiers — flagship (Pixel 6, Tensor G1), mid-range (Samsung Galaxy A54, Exynos 1380), and budget (Mediatek Helio G85) — the engine delivers p50 explanation latency under 40 ms and RSS overhead under 12 MB. Explanation artifacts are stored in `EncryptedSharedPreferences`-backed audit logs for regulatory inspection without third-party data transmission. The architecture satisfies HIPAA's minimum necessary standard, GDPR Article 22's right to explanation, EU AI Act Article 13's transparency requirement, and PCI-DSS scope containment for payment AI.

1. Introduction: The Interpretability Compliance Gap

The regulatory landscape for high-stakes AI has shifted materially in the past 24 months. The EU AI Act, effective August 2024, classifies AI systems used in medical devices, critical infrastructure, employment, and financial services as high-risk and mandates under Article 13 that they be "sufficiently transparent" to enable human oversight. HIPAA's minimum necessary standard has been interpreted by OCR enforcement to apply to model inputs when they contain Protected Health Information — a growing category as clinical AI on Android medical tablets proliferates. Card network rules and PCI-DSS 4.0 apply to cardholder data handling, and regulators have begun treating feature vectors derived from payment data as within scope.

The dominant interpretability tools were not designed with these constraints in mind. SHAP (Lundberg & Lee, 2017) in its reference implementation runs as a Python server; commercial wrappers — AWS SageMaker Clarify, Google Vertex Explainable AI, Azure Responsible AI Dashboard — are cloud services. All require the calling application to transmit input data to a remote endpoint. For a mobile application running a clinical decision support model, this means every explanation request sends a vector of lab values, vitals, and diagnosis codes to a third-party server. Under HIPAA, this is a disclosure of PHI requiring a Business Associate Agreement with the interpretability vendor. Obtaining and maintaining BAAs with cloud interpretability services is operationally complex and, for many Android-native deployments, has simply not happened — creating an unacknowledged compliance gap.

This paper documents a different architecture: an attribution engine that runs entirely within the Android application process, produces explanation artifacts locally, and creates no data transmission events at the interpretability tier. The compliance gap is closed by architectural elimination rather than contractual coverage.

2. Architecture: On-Device Attribution Engine

2.1 Layer Structure

The engine is organized as four layers, each with a distinct responsibility boundary:

Inference Layer. The engine integrates with TensorFlow Lite 2.13+, MediaPipe Tasks, and ONNX Runtime for Android via a common `InterpretabilityInterpreter` adapter interface. The adapter wraps the native interpreter and exposes a `runWithAttribution(input: TensorBuffer): AttributionResult` suspend function. The adapter accesses intermediate layer activations and input gradients through the framework's introspection APIs without modifying model weights or graph structure.

Interpretability Engine. Attribution runs as a structured Kotlin coroutine launched within a `SupervisorJob`-scoped `CoroutineScope`. The attribution algorithm (Section 2.2) operates on a pre-allocated `FloatArray` attribution buffer sized to the model's input feature count at initialization time. Pre-allocation eliminates GC pressure during attribution by avoiding heap allocation in the hot path. The coroutine is cancelled on scope disposal; there are no retained references to input tensors after the suspend function returns.

Memory Safety Layer. A watchdog coroutine polls `android.os.Debug.MemoryInfo` at 500 ms intervals, tracking RSS, native heap, and dalvik heap. When RSS exceeds a configurable threshold (default: 80% of `ActivityManager.getMemoryClass()` in MB converted to bytes), the watchdog sets a `AtomicBoolean` degradation flag. When the flag

is set, `runWithAttribution` returns a `AttributionResult.Degraded` containing the most recent cached partial attribution rather than computing a new one. This prevents the OOM killer from terminating the process mid-attribution, which would produce an unrecoverable state with no explanation artifact in the audit log.

Fault Isolation Layer. Each attribution request executes within its own `coroutineScope {}` block nested inside the `SupervisorJob` scope. A failure in one request's coroutine — an `OutOfMemoryError` caught as `Throwable`, an `IllegalStateException` from the inference framework, a timeout — does not cancel sibling coroutines or the parent scope. The `AttributionException` sealed class hierarchy distinguishes `OomDegradation`, `InferenceFrameworkError`, `TimeoutExpired`, and `ModelIncompatible` for structured error handling in the calling application.

2.2 Attribution Algorithm

Full Shapley value computation (Lundberg & Lee, 2017) requires evaluating the model on exponentially many feature subsets. For a model with n input features, exact SHAP is $O(2^n)$ — intractable on mobile hardware for any non-trivial feature count.

The engine implements Integrated Gradients (Sundararajan et al., 2017) as a gradient-based approximation. For an input x and a baseline x' (typically the zero vector or feature mean), the attribution for feature i is:

$$\text{IG}_i(x) = (x_i - x'_i) \int_{\alpha=0}^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} d\alpha$$

The integral is approximated with 50 Riemann steps — sufficient for <1% error on tabular models and far within the compute budget of mid-range Android hardware. Each step requires one forward pass and one backward pass through the model; the backward pass accesses gradients via TFLite's `SignatureDef` gradient API or ONNX Runtime's gradient session.

Rank correlation against full Shapley values (computed offline on the same models) is >0.94 across the three benchmark device configurations, measured on 500 randomly sampled test inputs. This is sufficient for operator-facing explanations and regulatory audit purposes, where relative feature importance ordering is the operative signal.

2.3 Why Kotlin Coroutines Over Java Thread Pools

Android applications that use `Executors.newFixedThreadPool()` or `AsyncTask` (deprecated) for long-running background operations frequently exhibit memory leaks because thread pool threads hold strong references to submitted `Runnable` objects, which in turn may hold references to `Activity` or `Context` objects. When an `Activity` is destroyed mid-computation, the `Runnable` prevents garbage collection of the entire view hierarchy.

Kotlin structured concurrency eliminates this class of leak. Coroutines launched within a `CoroutineScope` tied to a `ViewModel` or `LifecycleOwner` are automatically cancelled when the scope is destroyed. The coroutine's continuation object — the only live reference to computation state — is released immediately on cancellation. For a long-running interpretability library integrated into many host applications with varying lifecycle management discipline, this is a correctness guarantee that thread pools cannot provide.

3. Memory Safety on Android

3.1 Android Memory Model

Android runs applications as isolated processes forked from `Zygote`. Each app process has an independent heap managed by the ART runtime and a native heap for JNI allocations. The Android kernel uses a Low Memory Killer (LMK) that terminates processes when device memory pressure reaches defined thresholds, ordered by `oom_adj` score. Foreground application processes have the lowest `oom_adj` and are last to be killed, but they are not immune — a foreground process allocating beyond available physical memory can be killed with no warning.

The `ActivityManager.getMemoryClass()` API returns the per-application heap size limit in MB set by the device manufacturer (typically 256 MB on mid-range devices, 512 MB on flagship). This limit applies to the Dalvik/ART heap; native allocations via JNI can exceed it until the LMK intervenes at the system level.

3.2 Limitations of `MemoryInfo` Polling

`android.os.Debug.MemoryInfo` is the standard API for querying process memory usage. It provides PSS (Proportional Set Size), USS (Unique Set Size), RSS (Resident Set Size), and split breakdowns by allocation type. However, `getMemoryInfo()` is a blocking IPC call that takes 10–30 ms on some devices. Polling at high frequency to detect memory pressure in real time imposes its own CPU and latency overhead.

The watchdog coroutine resolves this tradeoff by polling at 500 ms intervals — slow enough to be negligible overhead, fast enough to detect a memory pressure trend before it becomes an OOM event. The watchdog does not attempt to prevent memory growth; it observes growth and sets a flag that causes the attribution engine to return cached results rather than allocating new buffers. Paired with `Runtime.getRuntime().gc()` called once when the degradation threshold is crossed, this typically recovers several MB of heap before the LMK becomes involved.

3.3 GC Pressure Metrics

GC pressure is measured via `Runtime.getRuntime().totalMemory()` - `Runtime.getRuntime().freeMemory()` sampled before and after attribution requests. In steady-state operation with the pre-allocated attribution buffer, GC pause events attributable to the interpretability engine are negligible — fewer than one GC event per 200 attribution requests in benchmarking. Without pre-allocation, attribution would allocate a `FloatArray` of size `n_features` per request, triggering minor GC on every call at scale.

4. Compliance Architecture

4.1 HIPAA

HIPAA's Privacy Rule (45 CFR §164.502) requires that covered entities and their business associates disclose PHI only as necessary for the permitted purpose. When a clinical AI model's input features are derived from PHI (lab results, vitals, ICD codes), transmitting those features to a remote SHAP server constitutes a disclosure. The receiving server is a business associate, requiring a BAA under 45 CFR §164.308(b).

The on-device architecture eliminates this disclosure. The attribution engine is a software library executing within the same process as the host application. There is no data transmission; there is no business associate relationship for the interpretability tier.

Audit logs are written to Android's `EncryptedSharedPreferences` using AES-256-GCM (the default cipher in Jetpack Security 1.1+). The audit record includes: input feature hash (SHA-256, not reversible to the original feature vector), attribution vector, confidence interval, model version, timestamp (UTC), and device ID. This log satisfies the HIPAA Security Rule's audit control requirement (45 CFR §164.312(b)) without storing raw PHI in the log itself.

4.2 GDPR Article 22

GDPR Article 22 grants data subjects the right not to be subject to solely automated decisions that produce legal or similarly significant effects. When Article 22(3) applies, controllers must implement "suitable measures to safeguard the data subject's rights and freedoms and legitimate interests," including "at least the right to obtain human intervention on the part of the controller, to express his or her point of view and to contest the decision." Recital 71 specifies this includes providing "meaningful information about the logic involved."

On-device attribution produces per-decision feature importance vectors that constitute the "meaningful information about the logic involved" required by Recital 71. Because the artifacts are generated locally and stored in the audit log, they are available immediately for subject access requests without querying a third-party service.

4.3 EU AI Act Article 13

Article 13(1) of the EU AI Act requires that high-risk AI systems "be designed and developed in such a way to ensure that their operation is sufficiently transparent to enable deployers to interpret the system's output and use it appropriately." Article 13(3)(b) specifies that the instructions for use must include "the level of accuracy, robustness and cybersecurity... against which the high-risk AI system has been tested and validated."

The attribution engine's per-prediction confidence intervals (derived from variance across the 50 Integrated Gradients steps) provide a machine-readable accuracy indicator for each explanation. The audit log structure is defined in the library's OpenAPI-compatible schema, enabling deployers to build compliance dashboards without reverse-engineering the output format.

4.4 PCI-DSS

PCI-DSS 4.0 Requirement 3 governs the protection of stored account data; Requirement 4 governs transmission of cardholder data over open networks. For fraud-scoring models running on Android POS terminals, the input feature vector may include merchant category codes, transaction amounts, velocity windows, and device identifiers derived from cardholder transaction history. Whether these constitute "cardholder data" under PCI definitions is a legal determination, but the practical risk is clear: transmitting them to a remote explanation server creates a data pathway that auditors will scrutinize.

On-device attribution removes this pathway from PCI scope assessment. There is no transmission of payment-adjacent features to a third party during explanation generation. PCI QSAs reviewing the system can confirm that the interpretability tier is entirely within the existing PCI boundary of the terminal.

5. Performance Benchmarking

5.1 Methodology

Benchmarks were run on three device configurations representing the primary Android deployment tiers for enterprise AI:

- **Flagship:** Pixel 6 (Google Tensor G1 SoC, 8 GB RAM, Android 13)
- **Mid-range:** Samsung Galaxy A54 (Exynos 1380, 6 GB RAM, Android 13)

- **Budget:** Xiaomi Redmi 10C (MediaTek Helio G85, 4 GB RAM, Android 12)

The model under test is a 128-feature tabular classification model (MobileNetV2-equivalent parameter count, fully-connected layers) exported as a TFLite FlatBuffer. Each device ran 1,000 attribution requests in sequence; p50 and p99 latency are reported from the full distribution. RSS overhead is measured as the delta in `Debug.MemoryInfo.totalPss` before library initialization and at steady state after 100 warm-up requests.

5.2 Results

Metric	Pixel 6 (Flagship)	Galaxy A54 (Mid-range)	Redmi 10C (Budget)
Explanation latency p50	22 ms	38 ms	61 ms
Explanation latency p99	71 ms	118 ms	204 ms
RSS overhead	9.4 MB	10.8 MB	11.6 MB
GC events / 100 requests	0.1	0.3	0.4
SHAP rank correlation	0.951	0.951	0.951

5.3 Remote SHAP Baseline Comparison

A reference remote SHAP server (self-hosted, same network as the device, 10 ms median network RTT) adds 180 ms median network latency to the above figures and fails with a network error at 0% connectivity. The on-device engine's p50 advantage over remote SHAP on mid-range hardware is approximately 142 ms under ideal network conditions. Under real-world mobile network conditions (LTE with intermittent packet loss), the advantage is larger. Under offline conditions, the remote alternative produces no explanation at all.

6. Integration Patterns

The engine exposes a single entry point:

`InterpretabilityInterpreter.wrap(nativeInterpreter, config)`. The `config` object specifies the baseline strategy (zero vector, feature mean, or custom), the number of integration steps (default: 50), the memory threshold percentage (default: 80), and the audit log path.

The wrapped interpreter's `runWithAttribution(input)` is a `suspend fun` returning `AttributionResult`, a sealed class with variants `Success`, `Degraded`, and `Error`. Callers are expected to handle all three variants; the library does not throw from the primary API path.

Detailed integration guides for TensorFlow Lite, MediaPipe Tasks, and ONNX Runtime are provided in `docs/integration/`.

7. Conclusion

Cloud-based AI interpretability imposes a structural compliance liability on mobile AI deployments handling regulated data. The liability is architectural: it cannot be patched with better data governance policies or vendor contracts alone, because the data transmission is inherent to the design of remote explanation services.

The on-device attribution engine described in this paper eliminates the liability at the source. By running gradient-based attribution inside the application process, storing explanation artifacts locally, and enforcing memory safety via a watchdog-and-degradation pattern, the engine delivers interpretability that is simultaneously compliant, low-latency, and offline-capable.

The library is available at `github.com/cryptofixyup/clear-code/tree/main/android-ai-interpretability`. Enterprise pilot inquiries, integration support, and compliance architecture reviews are available via the repository's enterprise pilot issue template.

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